

AMSC460: Homework 5

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4. The classical way to invert a matrix A in a basic linear algebra course augments A by the $n \times n$ identity matrix I and applies the Gauss–Jordan algorithm of Exercise 2 to this augmented matrix (including the solution part, i.e., the division by the pivots $a_{kk}^{(n-1)}$). Then A^{-1} shows up where I initially was.

How many floating point operations are required for this method? Compare this to the operation count of $\frac{8}{3}n^3 + \mathcal{O}(n^2)$ required for the same task using LU decomposition (see Example 5.5).

Proof. For the i -th row, we essentially need to zero out the other $n - 1$ row values in the same column. For each of these $n - 1$ subsequent rows, we need to multiply/divide the j -th row by a scalar, and then add/subtract from the rows, and we need to do this for both the original matrix and the new matrix which will become the inverse.

$$\sum_{i=1}^n (n - i + 1 + n - i + 1 + n + n + 1 + n)(n - 1) + 2n = + \sum_{i=1}^n (5n - 2i + 2)(n - 1) + 2n$$

where, to clarify,

- in the first parenthetical expression, suppose we are at the i -th row, and we want to zero out all terms below it. Consider the j -th row, $j > i$. Then we need to multiply the remaining $n - i$ nonzero elements in the j -th row by some scalar.
- Now add/subtract the scaled $n - i + 1$ values elementwise to the current i -th column
- In the soon-to-be identity, we cannot ignore previous columns, and so the multiplications and additions/subtractions must be done to all n elements in each of the rows, so another $n + n$
- The last 1 represents dividing by the pivot, and the n represents this same row transformation to the soon-to-be inverse
- Finally, for each row, take the dot product of the newfound row of the inverse by $\mathbf{b}$ to get x_i in $\mathbf{x}$, our solution vector

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- Repeat this process for all necessary $n - i$ rows, and then sum from 1 to n

$$\begin{aligned}
 &= \sum_{i=1}^n (5n - 2i + 2)(n - 1) + 2n \\
 &= \sum_{i=1}^n 5n^2 - 5n - 2in + 2i + 2n - 2 + 2n
 \end{aligned}$$

Simplifying like terms,

$$= \sum_{i=1}^n 5n^2 - 2in - n + 2i - 2$$

Breaking up the summation

$$\begin{aligned}
 &= 5n^3 - n^2 - 2n - (2n - 2) \left(\sum_{i=1}^n i \right) \\
 &= 5n^3 - n^2 - 2n - (2n - 2) \frac{n(n+1)}{2} \\
 &= 4n^3 + \mathcal{O}(n^2)
 \end{aligned}$$

which is far slower than the LU decomposition method. □

5. Let

$$A = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 0 & 4 & 3 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & -2 \end{pmatrix}.$$

(a) The matrix A can be decomposed using partial pivoting as

$$PA = LU,$$

where U is upper triangular, L is unit lower triangular, and P is a permutation matrix. Find the 4×4 matrices U , L , and P .

(b) Given the right-hand-side vector $\mathbf{b} = (26, 9, 1, -3)^T$, find $\mathbf{x}$ that satisfies $A\mathbf{x} = \mathbf{b}$. (Show your method: do not just guess.)

Proof.

(a) Let $P^{(1)}$ and $M^{(1)}$ be the I_4 , as there is no need for partial pivoting in the first column or zero'ing out the first column. Thus, $A^{(1)} = M^{(1)}P^{(1)} = A$. Similarly, for the second column, we have $A^{(2)} = M^{(2)}P^{(2)}M^{(1)}P^{(1)} = A$. In the third column, we must switch the third and the fourth row, such that

$$P^{(3)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$M^{(3)} = I_4$ since there is already a 0 in the below entry due to the swapping of the rows done by the permutation matrix. Thus,

$$U = M^{(3)}P^{(3)}M^{(2)}P^{(2)}M^{(1)}P^{(1)}A = P^{(3)}A = \begin{bmatrix} 5 & 6 & 7 & 8 \\ 0 & 4 & 3 & 2 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Given all of the permutation matrices, we can now calculate P which is

$$P = P^{(3)}P^{(2)}P^{(1)} = P^{(3)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Finally, since $L = P(M^{(3)}P^{(3)}M^{(2)}P^{(2)}M^{(1)}P^{(1)})^{-1}$, this nicely simplifies to

$$L = P^{(3)}P^{(3)} = I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(b) Note that $P = P^{(3)}$ and that $(P^{(3)})^{-1} = P^{(3)}$, observe that

$$PA = LU \implies P^{-1}PA = P^{-1}LU \implies A = PLU$$

Thus, we want to find $\mathbf{x}$ such that $PLU\mathbf{x} = \mathbf{b} = (26, 9, 1, -3)^T$. Note that

$$PLU\mathbf{x} = \mathbf{b} \implies LU\mathbf{x} = P\mathbf{b} = (26, 9, -3, 1)^T$$

Now, we can proceed with forward and backward substitution. Let $\mathbf{y} = U\mathbf{x}$. We seek $\mathbf{y}$ such that

$$L\mathbf{y} = (26, 9, -3, 1)^T \implies \mathbf{y} = (26, 9, -3, 1)^T$$

because $L = I_4$. Now we seek the original $\mathbf{x}$ such that

$$U\mathbf{x} = \begin{bmatrix} 5 & 6 & 7 & 8 \\ 0 & 4 & 3 & 2 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 26 \\ 9 \\ -3 \\ 1 \end{bmatrix}$$

By backward substitution, $x_4 = 1 \implies -x_3 - 2 = -3 \implies x_3 = 1$, and continuing in this manner yields

$$\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

□

6. Let B be any real, nonsingular $n \times n$ matrix, where n is even, and set $A = B - B^T$. Show that A does not admit an LU decomposition (i.e., some pivoting *must* be applied, even if A is nonsingular).

Proof. Note that $A = B - B^T$ omits the following form

$$A = \begin{bmatrix} 0 & a & b & \cdots & A_{1n} \\ -a & 0 & c & \cdots & A_{2n} \\ -b & -c & 0 & \cdots & A_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -A_{1n} & -A_{2n} & -A_{3n} & \cdots & 0 \end{bmatrix}$$

where this is known as a **skew symmetric matrix**, which have the property that $A^T = -A$. Furthermore, recall that if a square matrix X is invertible, then $\det(A) = \det(A^T)$. We will show that a skew symmetric square matrix $X \in \mathbb{R}^{n' \times n'}$, where n' is odd, must have $\det(X) = 0$.

$$X^T = -X \implies \det(X^T) = \det(-X) \implies \det(X^T) = (-1)^{n'} \det(X)$$

This last equation can only be satisfied if $\det(X) = \det(X^T) = 0$. Thus, any odd-sized square symmetric matrix is singular. In the context of our problem, while A need not be singular, its odd-sized principal minors – the smaller square submatrices – must be singular, and so there does not exist an LU decomposition of A without pivoting. $\square$